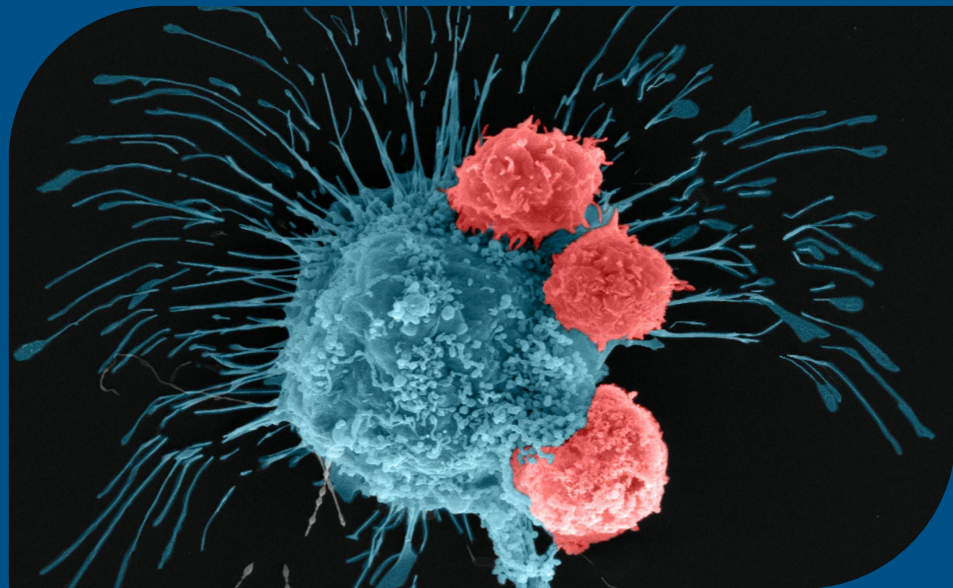


Wisconsin Breast Cancer

MCC Data Analytics Applications
and Modeling
Michael Hodgetts

Fernando Najera-Medina



Analysis Background

Dataset for Analysis

<https://media.geeksforgeeks.org/wp-content/uploads/20240522174620/data-2.csv>

Description of Dataset

The dataset contains features computed from digitized images of fine needle aspirate samples of breast mass tissue.

These features should help in categorizing breast tumor growths as either **malignant** or **benign**.

Where does this analysis fit in the diagnosis process?

Breast cancer diagnosis follows a highly standardized pathway known in the medical community as the Triple Assessment. This approach combines clinical evaluation, diagnostic imaging, and tissue analysis to ensure absolute diagnostic accuracy.

This analysis fits in the tissue analysis part, and its result should add relevant data that will ensure final diagnostic accuracy.

General feature description

1. mean radius: Mean of distances from center to points on the perimeter.
2. mean texture: Standard deviation of gray-scale values.
3. mean perimeter: Perimeter of the tumor.
4. mean area: Area of the tumor.
5. mean smoothness: Variation in radius lengths.
6. mean compactness: $\text{Perimeter}^2 / \text{Area} - 1.0$.
7. mean concavity: Severity of concave portions of the contour.
8. mean concave points: Number of concave portions of the contour.
9. mean symmetry: Symmetry of the cell nuclei.
10. mean fractal dimension: "Coastline approximation" - 1

Considerations prior to analysis

Malignant cells will tend to be larger, more irregularly shaped and chaotic than benign cells that tend to be small, evenly spaced and more smooth shaped

Objective

Create a binary classifier that will accurately predict if a new observation, with the same features as the previously described dataset, is either a malignant or benign tumor.

The classifier must perform perfectly when identifying true positives and must avoid any false negatives.

Exploratory Data Analysis

Dataset Description

Total Number of Records: 569

Number of Total Malignant Diagnosis Cases: 212, (37.0%)

Number of Total Non Malignant Diagnosis Cases: 357, (63.0%)

Number of feature columns: 30

All features are of float dtype.

The target variable is categorical and will need to be codified to binary prior to analysis.

There are 2 columns that are useless for this analysis: 'id', 'Unnamed: 32'

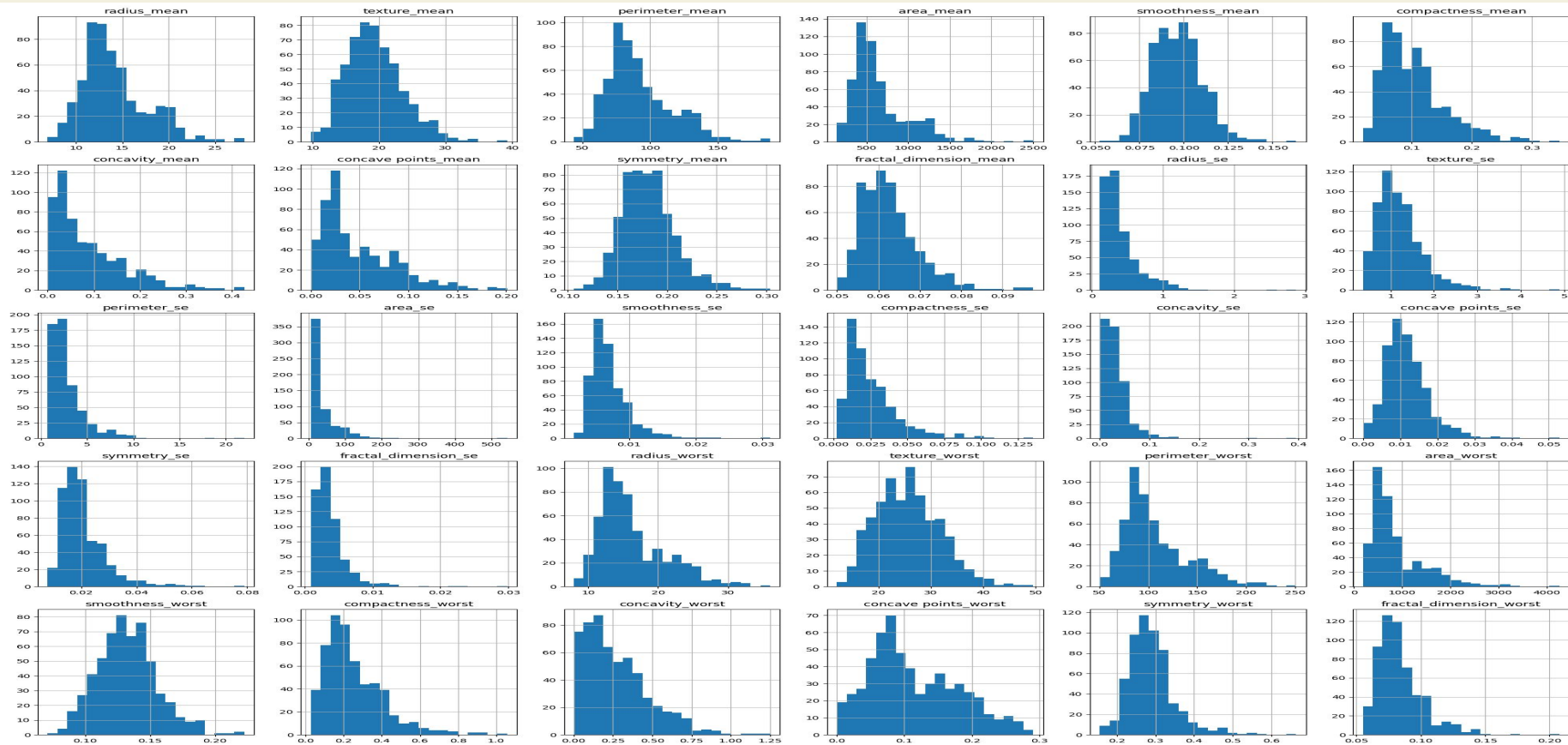
There are no missing fields in any row.

The sorted correlation of each numeric variable with the target variable (diagnosis) is:

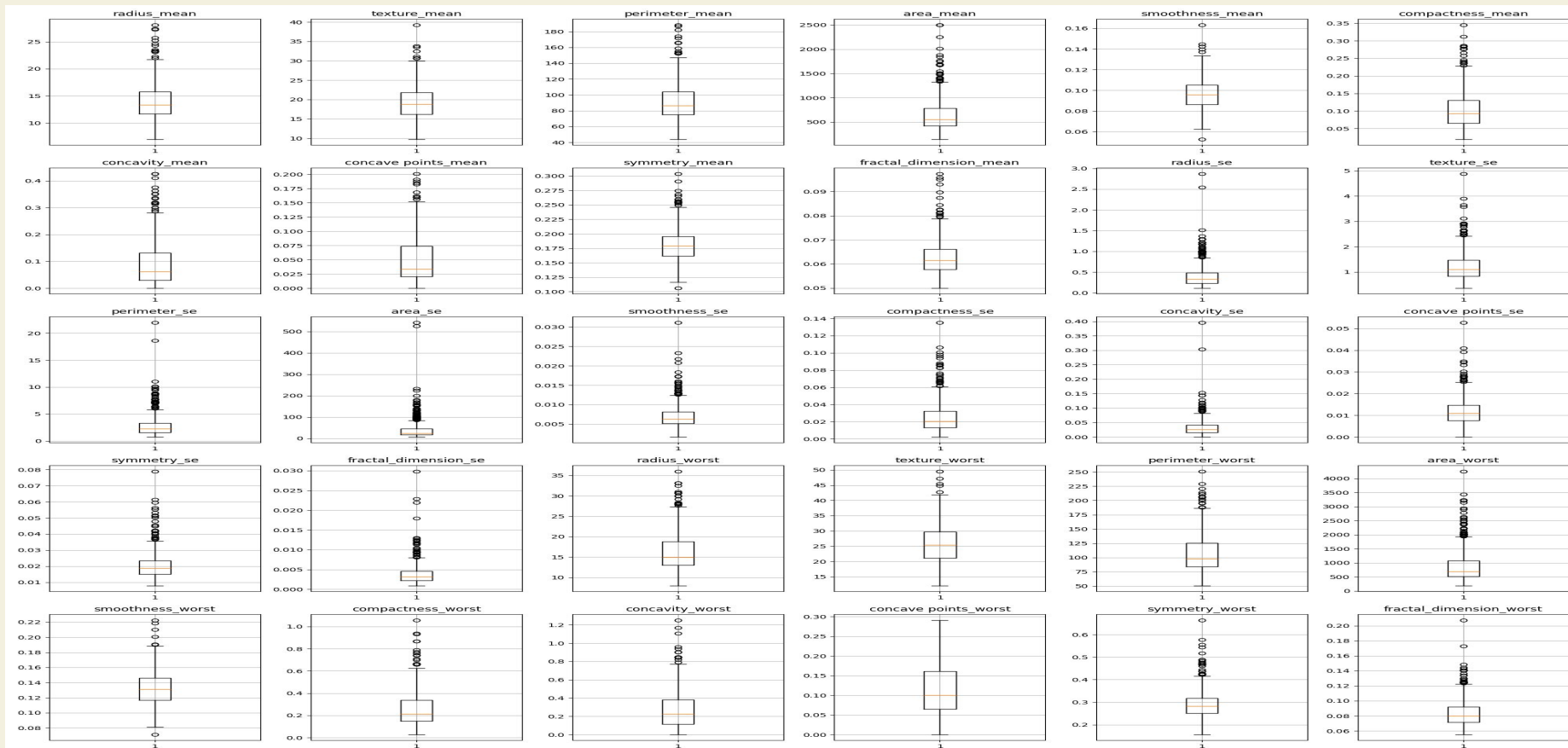
concave points_worst	0.793566	texture_worst	0.456903
perimeter_worst	0.782914	smoothness_worst	0.421465
concave points_mean	0.776614	symmetry_worst	0.416294
radius_worst	0.776454	texture_mean	0.415185
perimeter_mean	0.742636	concave points_se	0.408042
area_worst	0.733825	smoothness_mean	0.358560
radius_mean	0.730029	symmetry_mean	0.330499
area_mean	0.708984	fractal_dimension_worst	0.323872
concavity_mean	0.696360	compactness_se	0.292999
concavity_worst	0.659610	concavity_se	0.253730
compactness_mean	0.596534	fractal_dimension_se	0.077972
compactness_worst	0.590998	symmetry_se	-0.006522
radius_se	0.567134	texture_se	-0.008303
perimeter_se	0.556141	fractal_dimension_mean	-0.012838
area_se	0.548236	smoothness_se	-0.067016

Data Plots

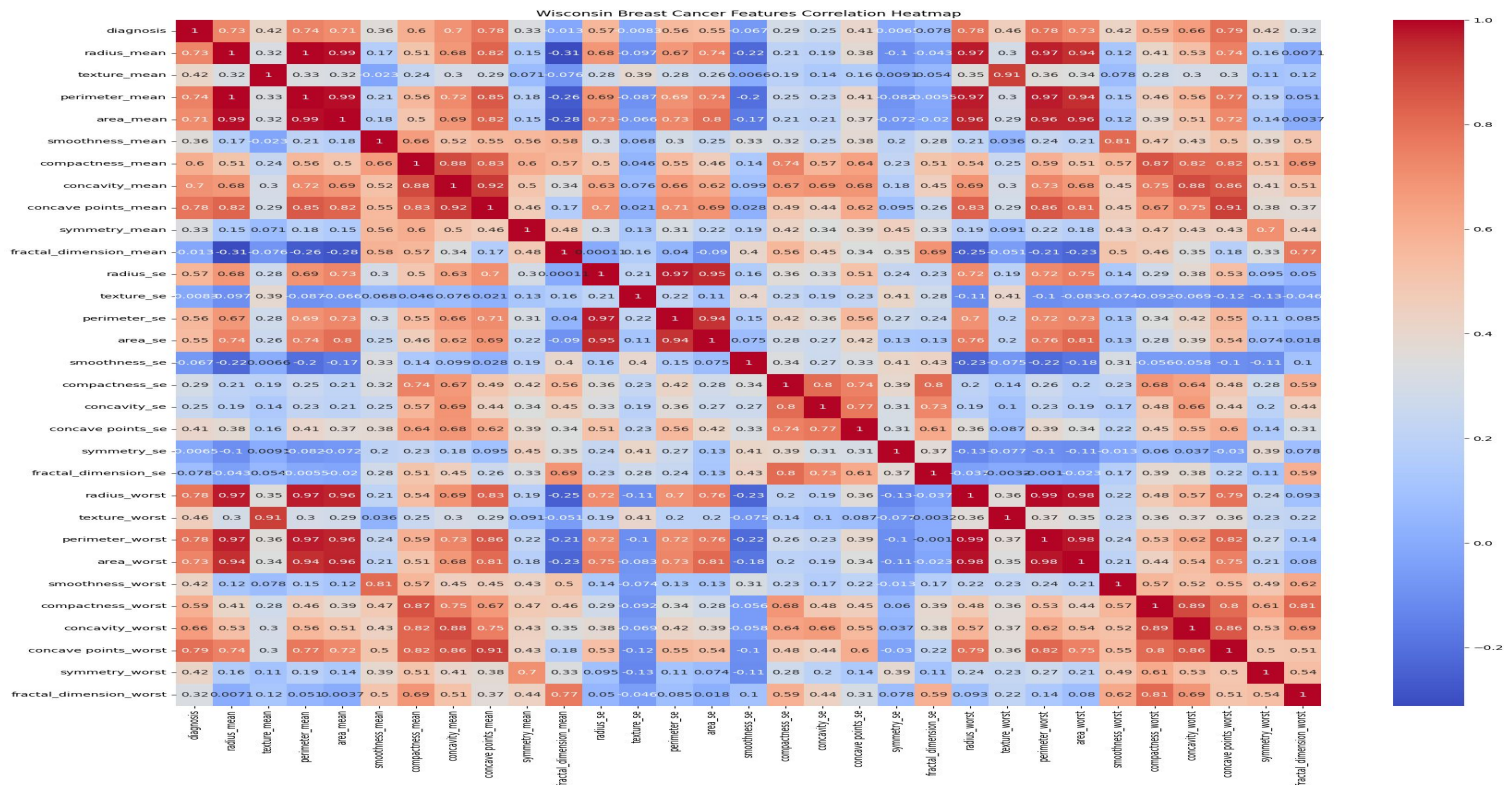
Histograms of Features



Box Plot of Features



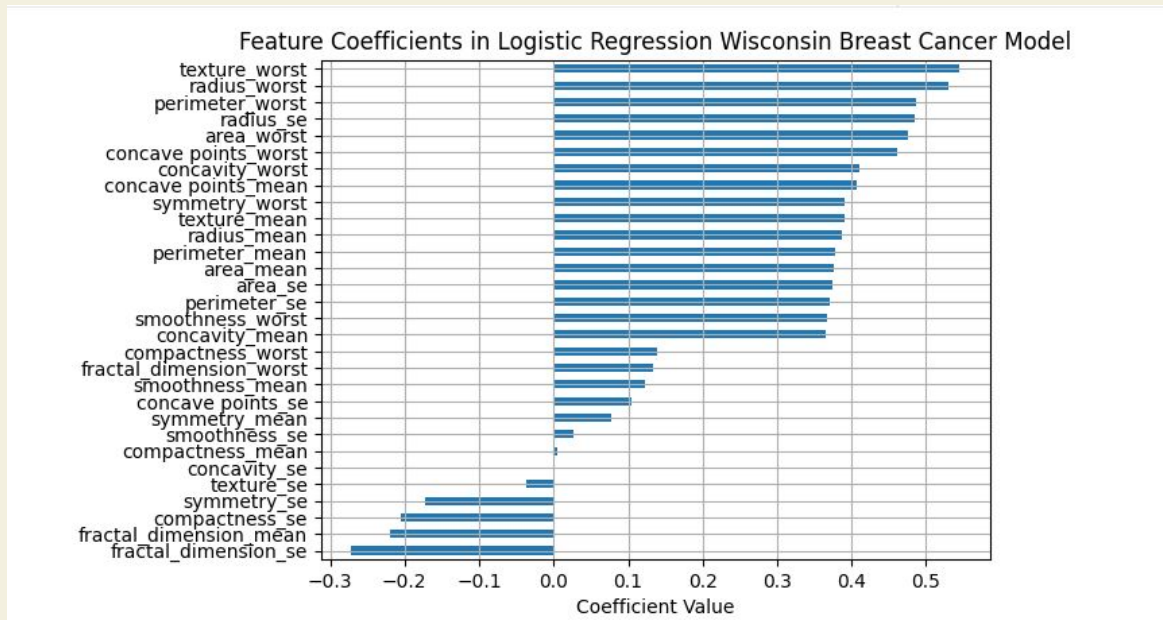
Correlation Matrix



Model and Performance Evaluation

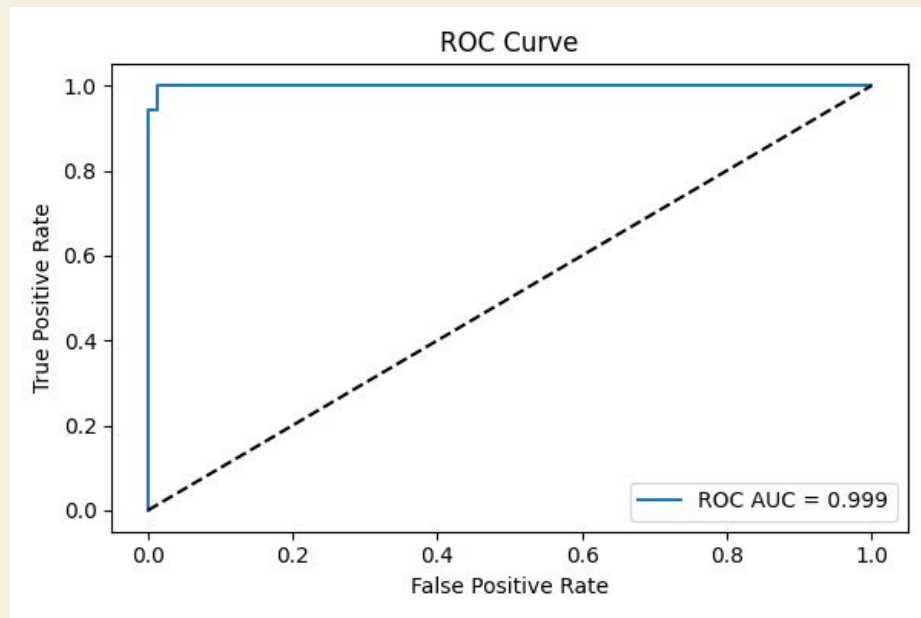
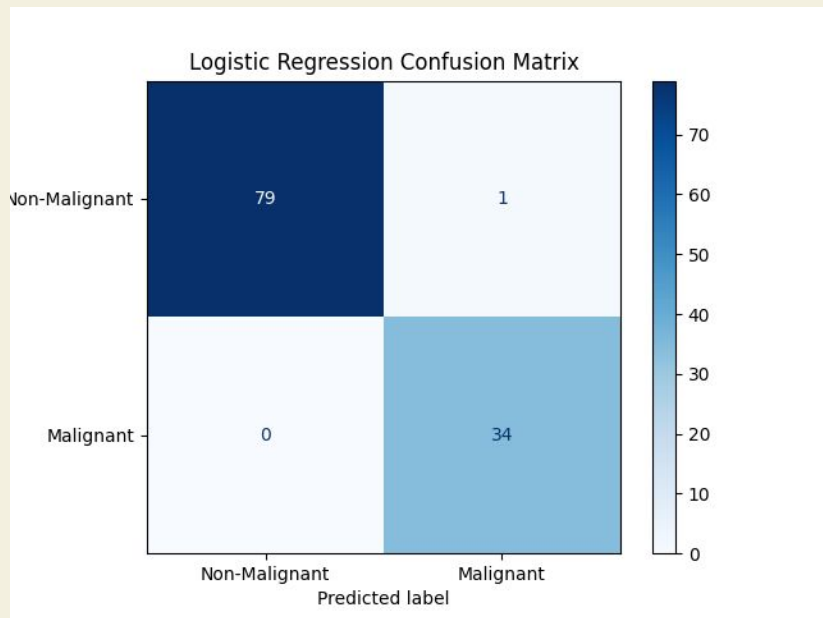
Logistic Regression Model

- The dataset was split %80 for training and %20 for testing.
- All features were scaled prior to model training.
- The **score** for the logistic regression model was: **0.9912280701754386**
- The **ROC-AUC** score was: **0.999**
- The following image shows the resulting coefficients for the features:



Confusion Matrix and ROC-AUC Curve

- Accuracy: 0.991
- Precision: 0.971
- Recall: 1.000
- F1: 0.986
- Roc_auc: 0.999
- Log loss: 0.09505042559397929



Further Testing

With a working model, focus now turns to optimization. For this purpose the general approach was to **reduce** the amount of training features.

I ran tests that used the top 10, top 15, and top 20 features, but the models performed poorly.

Upon inspection of the features, there seems to be categories of features. For example, there are 3 features in a 'concave' group:

- concave points_mean
- concave points_se
- concave points_worst

I am assuming that there might be a good amount of collinearity between these features so I'm hoping that using only the best performing one will still yield good results.

The features with the highest correlation to diagnosis per family are:

- concave points_worst
- perimeter_worst
- radius_worst
- area_worst
- compactness_mean
- texture_worst
- smoothness_worst
- symmetry_worst
- fractal_dimension_worst

Further Testing Results

“Non-collinear” Model

The performance scores for this new model were:

Accuracy: 0.982

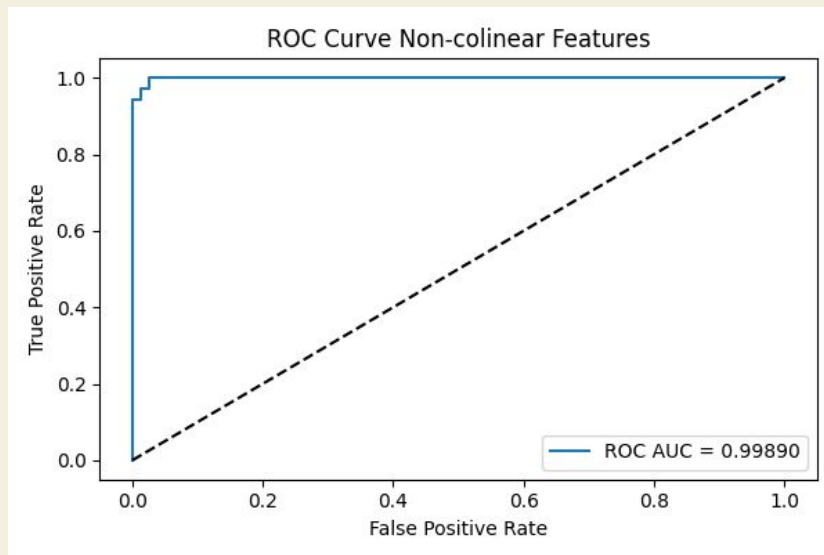
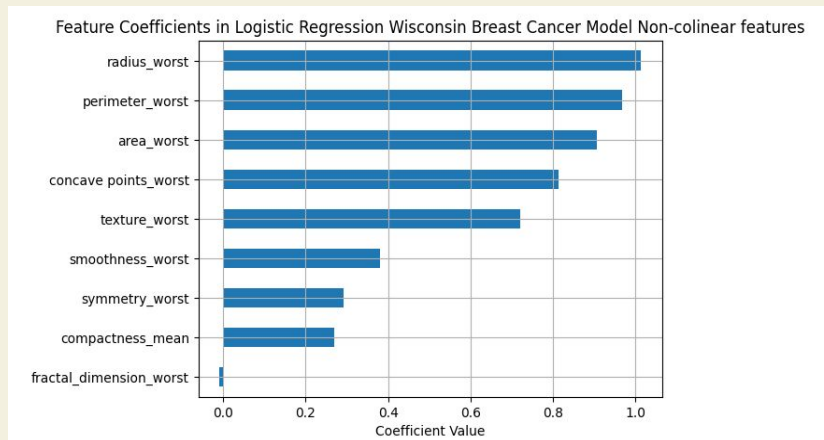
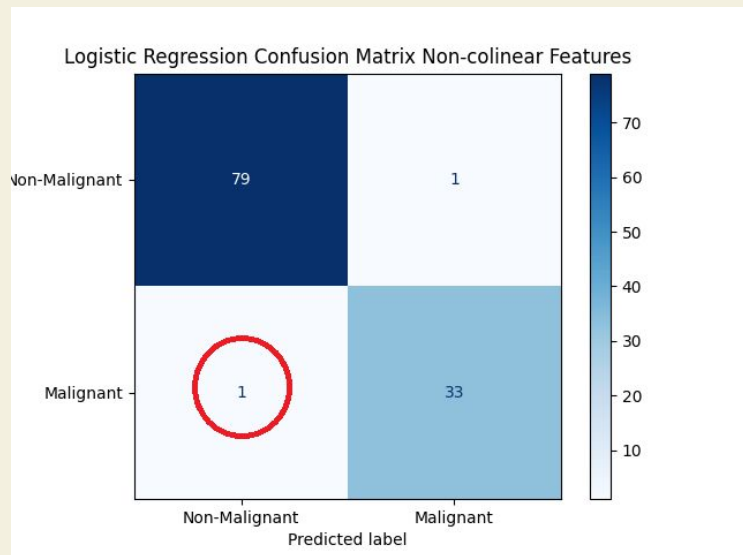
Precision: 0.971

Recall: 0.971

F1: 0.971

Roc_auc: 0.999

False_negatives: 1.000



Adjusting Threshold

“Non-collinear” with adjusted threshold Model

Accuracy: 0.982

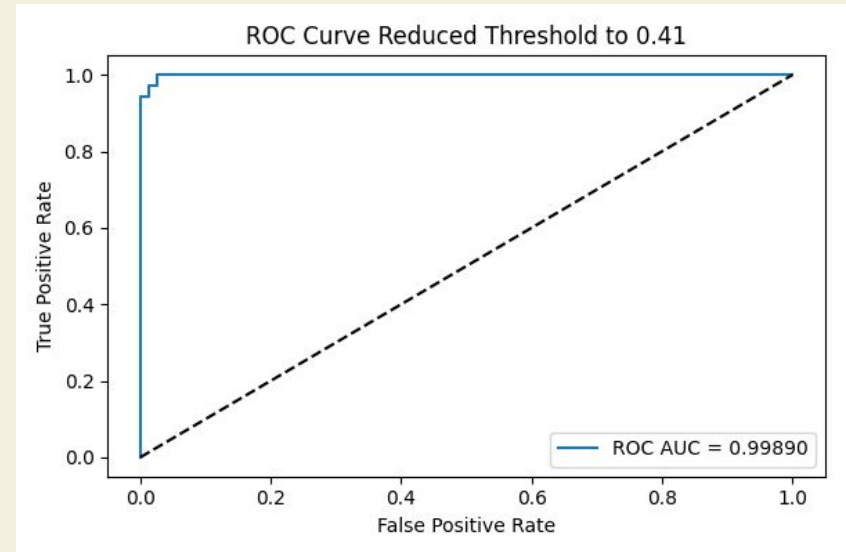
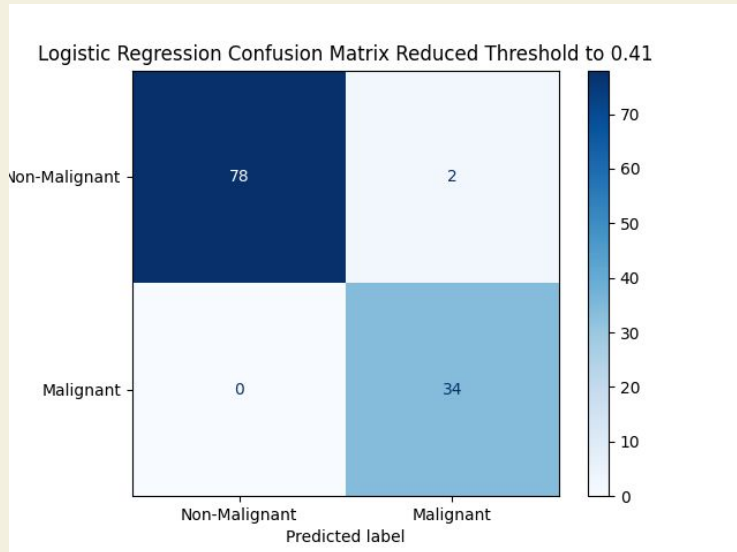
Precision: 0.944

Recall: 1.000

F1: 0.971

Roc_auc: 0.999

False_negatives: 0.000



Adding Features

- As an alternate path, I'll try adding features to the “non-collinear” set of features.
- I will add one feature at a time and stop when I have zero false negatives.
- I will select the features with the highest correlation to diagnosis.

Optimized Number of Features

Total features used: 14

Accuracy: 0.982

Precision: 0.944

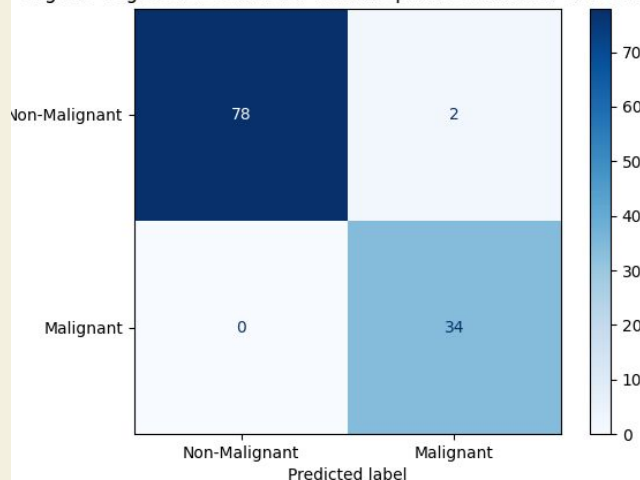
Recall: 1.000

F1: 0.971

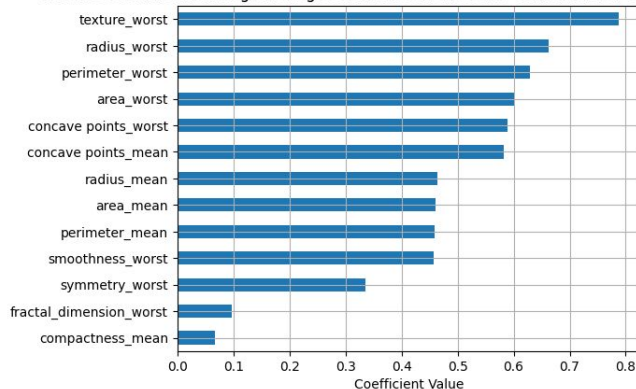
Roc_auc: 0.999

False_negatives: 0.000

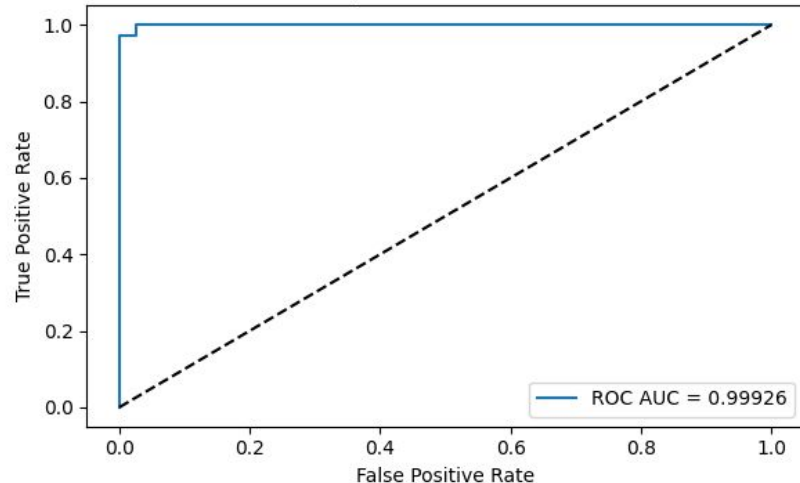
Logistic Regression Confusion Matrix Optimized number of Features



Feature Coefficients in Logistic Regression Wisconsin Breast Cancer Model Optimized Features



ROC Curve Optimized number of Features



Conclusions

Initial model:

- The initial model performs very well, accurately identifying 100% of all malignant instances. This is extremely desirable in breast cancer diagnosis since you want to make sure no critical cases were missed.
- There is a very small 2.9% chance of reporting a False Positive.
- With a Log Loss of 0.095, the model is highly confident about making correct choices.
- The ROC-AUC score of 0.999 indicates that the model is almost flawless when separating the two classes.

Secondary models:

- Both, the threshold adjusted, and the optimized feature models accurately identify %100 of all malignant instances.
- There is a slightly higher possibility to report false positives than the initial model.
- With fewer features than the initial model these models might be better suited for deployment.

Future considerations:

I believe the models can be put to further testing in an real world workflow to see how they perform against the existing diagnosis method. The goal would be to optimize further or deploy as is.

Thank You!